MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics **CBSE CLASS 10 – CHAPTER TEST (SET – 7)** **Time:** 45 Minutes
Class: Class X (CBSE) **Chapter 7: Coordinate Geometry** **MM: 20**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choices are provided wherever specified.
2. **Section A** comprises Multiple Choice / Objective Questions of 1 mark each.
3. **Section B** comprises Short Answer Type questions.
4. **Section C** comprises Long Answer & Competency Based questions.
5. Use of calculators or electronic devices is strictly prohibited. Draw neat figures wherever required.

SECTION A: Multiple Choice Questions (5 × 1 = 5 Marks)

Q1. The coordinates of the end points of a diameter of a circle are $(5, -2)$ and $(5, 2)$. The length of the radius of the circle is : [1]

(A) ± 2 (B) ± 4 (C) 4 (D) 2

Q2. The end points of a diameter of circle are $(2, 4)$ and $(-3, -1)$. The length of its radius is : [1]

(A) $\frac{5\sqrt{2}}{2}$ units (B) $5\sqrt{2}$ units (C) $3\sqrt{2}$ units (D) $\pm\frac{5\sqrt{2}}{2}$ units

Q3. In the given figure, a circle is centred at $(1, 2)$. The diameter of the circle is : [1]

(A) 4 (B) $2\sqrt{2}$ (C) $\sqrt{5}$ (D) $2\sqrt{5}$

Q4. A circle centred at $(-1, 2)$ passes through the point $(0, 3)$. Radius of the circle is : [1]

(A) $2\sqrt{2}$ (B) $\sqrt{2}$ (C) $\sqrt{26}$ (D) 1

Q5. A(-4, 5) and C(8, 2) are the two opposite vertices of a parallelogram ABCD. Its diagonals intersect each other at P(a, b). The relation between 'a' and 'b' is : [1]

(A) $b = a - 1.5$ (B) $b = a + 1.5$ (C) $b = a - 4.5$ (D) $b = a + 4.5$

SECTION B: Short Answer Type Questions (6 Marks)

Q6. If the point $C(k, 4)$ divides the line segment joining $A(2, 6)$ and $B(5, 1)$ internally in the ratio 2 : 3, find the value of k. [2]

Q7. Determine the ratio in which the line $3x + y - 9 = 0$ divides the line segment joining the points $(1, 3)$ and $(2, 5)$. Find the point of intersection. [2]

Q8. A point $P(x, 7)$ divides a line segment joining the points $A(-5, 4)$ and $B(7, 9)$ in a certain ratio. Find the ratio and hence find the value of x. [2]

SECTION C: Long Answer & Competency Based Questions (9 Marks)

Q9. A circle centered at $(2, 1)$ passes through the points $A(5, 6)$ and $B(-3, K)$. Find the value(s) of K. Hence find length of chord AB. ¿ **OR** ¿ (b) Prove that the point P dividing the line segment joining the points $A(-1, 7)$ and $B(4, -3)$ in the ratio $3 : 2$, lies on the line $x - 3y = -1$. Also find length of PA and PB. ¿ ¿ **[5]**

Q10. Competency Case Study: Two cellular transmission towers are situated at points $A(-3, 5)$ and $B(7, 9)$ on a regional map grid. A booster relay is installed at a point P along the direct cable route AB such that $AP : PB = 2 : 3$. **[4]**

(i) Find the coordinates of the booster relay P.

(ii) Find the total direct aerial distance between tower A and tower B.

(iii) If a third monitoring station is placed at $C(2, 7)$, determine whether station C is equidistant from tower A and tower B.

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